

Data Structure and Algorithm - 1

Technical Workouts

1. Understand Data Structure & Algorithms

- a. What is a Data Structure
- b. Linear vs Non-linear data structures
- c. Built-in vs Custom data structures
- d. What is an Algorithm
- e. Input, Processing and Output

2. Understand Memory Allocation & Memory Leak

- a. What is memory allocation
- b. Static vs Dynamic memory allocation
- c. Static vs Dynamic typing
- d. What is a memory leak
- e. Garbage collection

3. Understand Complexity Analysis

- a. Time complexity
- b. Space complexity
- c. Types of time complexity
- d. Constant, Linear, Quadratic and Cubic time
- e. Logarithmic, Linearithmic, Exponential and Factorial time

4. Understand Asymptotic Analysis

- a. What is asymptotic notation
- b. Big O notation (upper bound / worst case)
- c. Big Omega notation (lower bound / best case)
- d. Big Theta notation (tight bound)
- e. Comparing growth rates

5. Understand Recursion

- a. What is recursion
- b. Base case and recursive case
- c. Why recursion when loops exist
- d. Direct, Indirect, Tail and Head recursion
- e. Tree, Linear and Nested recursion

6. Understand Array

- a. Array characteristics
- b. Advantages and disadvantages of arrays
- c. Static vs Dynamic arrays
- d. Python list operations
- e. Tuple, Set, Dictionary and Shallow vs Deep copy

7. Understand Linear Search & Binary Search

- a. What is a searching algorithm
- b. Linear search
- c. Binary search
- d. Lower bound and upper bound
- e. $O(n)$ vs $O(\log n)$ comparison

8. Understand Linked List

- a. What is a linked list and its nodes
- b. Linked list operations
- c. Singly and Doubly linked lists
- d. Circular linked lists
- e. Slow and fast pointer concept

9. Understand String

- a. What is a string
- b. String operations
- c. Indexing and slicing
- d. Replacing characters in a string
- e. Sample workouts